

Lab 03: Real-World Data Linear Regression

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Part A. Data

1. Dataset source

Source: NASA Goddard Institute for Space Studies (GISS) – GISS Surface Temperature Analysis (GISTEMP), version 4.

Full URL: <https://data.giss.nasa.gov/gistemp/> (annual global mean table:
https://data.giss.nasa.gov/gistemp/tabledata_v4/GLB.Ts+dSST.csv)

2. Description of the data

GISTEMP is NASA's official record of global surface temperature change, produced from about 6,300 meteorological stations on land combined with sea-surface temperature measurements from ships, buoys, and satellites. The quantity reported is the annual global-mean temperature anomaly: the difference, in degrees Celsius, between the average surface temperature of a given year and the 1951–1980 baseline average. This lab uses the 20 most recent complete years of annual data, 2006–2025.

3. Variables

Independent variable x: Year (calendar year, dimensionless count of years)

Dependent variable y: Global mean temperature anomaly, in degrees Celsius (°C), relative to the 1951–1980 base period

4. Table of observations (n = 20)

Year (x)	Temperature Anomaly, deg C (y)
2006	0.6417
2007	0.6658
2008	0.5450
2009	0.6567
2010	0.7250
2011	0.6075
2012	0.6467
2013	0.6775
2014	0.7492
2015	0.8967
2016	1.0117
2017	0.9142
2018	0.8492
2019	0.9775
2020	1.0067
2021	0.8475
2022	0.8908
2023	1.1675
2024	1.2842
2025	1.1925

Part B. Python Script

The complete script below (also submitted separately as Lopez_Francine Joy_3H_Lab03.py) reads the data, computes the least-squares regression coefficients using the sums-of-squares formulas, computes the fit statistics, generates the plots, and makes a prediction.

```
"""
Lab 03: Real-World Data Linear Regression
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Data set : NASA GISS Surface Temperature Analysis (GISTEMP v4)
Source   : NASA Goddard Institute for Space Studies
URL      : https://data.giss.nasa.gov/gistemp/
          (annual global mean table: GLB.Ts+dSST.csv)

x = Year (calendar year)
y = Global mean surface temperature anomaly (deg C, relative to the
    1951-1980 base period)

The script:
1. Stores the 20 paired observations (2006-2025) used for the fit.
2. Computes the least-squares regression  $y = a_0 + a_1x$  "by hand"
   using the standard sums-of-squares formulas from class.
3. Computes  $S_r$  (SSE),  $r^2$  and the standard error of the estimate  $sy/x$ .
4. Plots the data with the fitted line, and a residual plot.
5. Uses the fitted line to predict  $y$  for a year outside the data set.
"""

import numpy as np
import matplotlib.pyplot as plt

# -----
# 1. Data
# -----
# x = Year, y = Global Land-Ocean Temperature Index anomaly (deg C)
x = np.array([2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015,
              2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025],
             dtype=float)

y = np.array([0.6417, 0.6658, 0.5450, 0.6567, 0.7250, 0.6075, 0.6467,
              0.6775, 0.7492, 0.8967, 1.0117, 0.9142, 0.8492, 0.9775,
              1.0067, 0.8475, 0.8908, 1.1675, 1.2842, 1.1925])

n = len(x) # number of paired observations

# -----
# 2. Least-squares fit:  $y = a_0 + a_1x$ 
# Using the standard sums-of-squares formulas
#  $a_1 = (n \cdot \text{Sum}(xy) - \text{Sum}(x) \cdot \text{Sum}(y)) / (n \cdot \text{Sum}(x^2) - (\text{Sum}(x))^2)$ 
#  $a_0 = \bar{y} - a_1 \bar{x}$ 
# -----
sum_x = np.sum(x)
sum_y = np.sum(y)
sum_xy = np.sum(x * y)
sum_x2 = np.sum(x ** 2)

x_bar = sum_x / n
y_bar = sum_y / n

a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
a0 = y_bar - a1 * x_bar

print(f"Number of observations, n = {n}")
print(f"Slope a1 = {a1:.6f} (deg C per year)")
print(f"Intercept a0 = {a0:.4f} (deg C)")
print(f"Regression equation:  $y = {a0:.4f} + {a1:.6f} * x$ ")

# -----
# 3. Goodness of fit:  $S_t$ ,  $S_r$ ,  $r^2$ , and standard error  $sy/x$ 
# -----
y_pred = a0 + a1 * x # values predicted by the line
residuals = y - y_pred # residuals (observed - predicted)

St = np.sum((y - y_bar) ** 2) # total sum of squares about the mean
Sr = np.sum(residuals ** 2) # sum of squares of residuals (SSE)

r2 = (St - Sr) / St # coefficient of determination
r = np.sqrt(r2) # correlation coefficient

# standard error of the estimate (n-2 degrees of freedom for a line fit)
syx = np.sqrt(Sr / (n - 2))

print(f"St (total SS) = {St:.6f}")
print(f"Sr (SSE) = {Sr:.6f}")
print(f" $r^2 = {r2:.6f}$ ")
print(f" $r = {r:.6f}$ ")
print(f"Standard error  $sy/x = {syx:.6f}$  deg C")
```

```

# -----
# 4a. Plot: data points with the fitted line
# -----
plt.figure(figsize=(7, 5))
plt.scatter(x, y, color="tab:blue", label="Observed data")
x_line = np.linspace(x.min(), x.max(), 100)
plt.plot(x_line, a0 + a1 * x_line, color="tab:red",
         label=f"Fit: y = {a0:.3f} + {a1:.5f}x")
plt.xlabel("Year")
plt.ylabel("Global temperature anomaly, deg C (vs. 1951-1980)")
plt.title("Global Temperature Anomaly vs. Year (GISTEMP, 2006-2025)")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig("fit_plot.png", dpi=200)
plt.close()

# -----
# 4b. Residual plot
# -----
plt.figure(figsize=(7, 5))
plt.scatter(x, residuals, color="tab:green")
plt.axhline(0, color="black", linewidth=1)
plt.xlabel("Year")
plt.ylabel("Residual, deg C (observed - predicted)")
plt.title("Residual Plot")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig("residual_plot.png", dpi=200)
plt.close()

# -----
# 5. Prediction for a year not in the data set
# -----
x_new = 2030.0
y_new = a0 + a1 * x_new
print(f"Prediction for x = {x_new:.0f}: y = {a0:.4f} + "
      f"{a1:.6f}*{x_new:.0f} = {y_new:.4f} deg C")

```

Part C. Regression Results

1. Regression equation

$$y = a_0 + a_1 x$$

$$y = -62.9258 + 0.031642 x \quad (y \text{ in } ^\circ\text{C}, x = \text{calendar year})$$

2. Coefficients

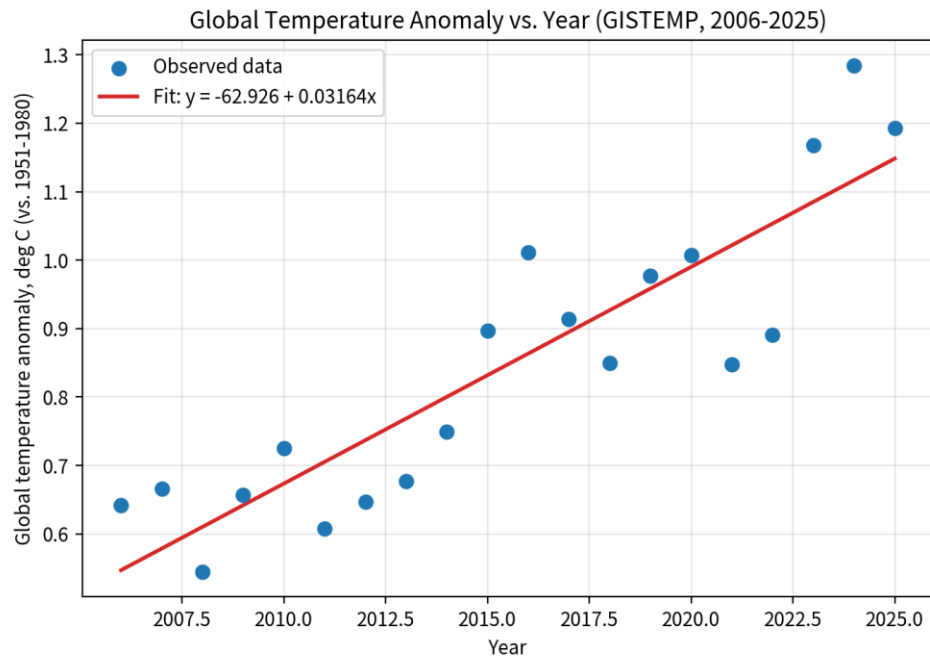
Slope, $a_1 = 0.031642$ °C per year

Intercept, $a_0 = -62.9258$ °C

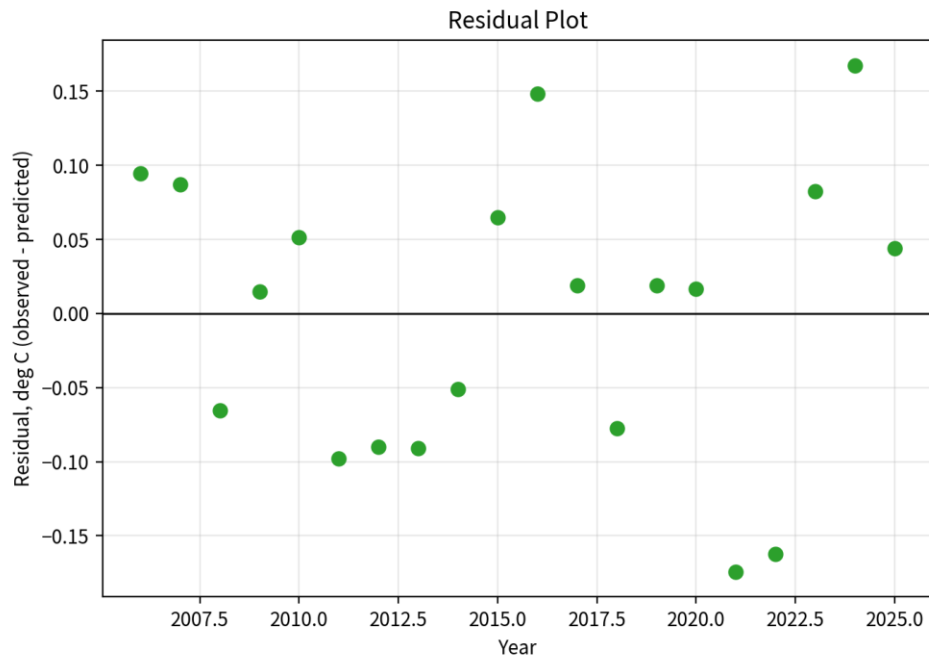
3. Fit statistics

Quantity	Value
St (total sum of squares)	0.844854
Sr (SSE, sum of squares of residuals)	0.179066
r^2 (coefficient of determination)	0.7881
r (correlation coefficient)	0.8877
sy/x (standard error of the estimate)	0.0997 °C

4. Graph of data with fitted line



5. Residual plot



6. Prediction

Predict the anomaly for $x = 2030$ (a year not in the data set):

$$y = -62.9258 + 0.031642 \times (2030) = -62.9258 + 64.2323 = 1.3065 \text{ }^{\circ}\text{C}$$

Predicted global mean temperature anomaly for 2030 $\approx 1.31 \text{ }^{\circ}\text{C}$ above the 1951–1980 baseline. This prediction is meaningful because it extrapolates the observed 20-year warming trend only 5 years beyond the data set (a modest extrapolation), giving a data-driven estimate that can be compared against climate-model projections and used to gauge how close the planet is tracking toward international warming targets (such as the $1.5 \text{ }^{\circ}\text{C}$ threshold referenced in the Paris Agreement). As with any extrapolation, it assumes the recent linear trend continues and does not account for possible acceleration, volcanic cooling events, or strong El Niño/La Niña swings.

7. Interpretation

Slope: $a_1 = 0.0316 \text{ }^{\circ}\text{C}/\text{year}$ means the global mean surface temperature anomaly has been rising, on average, by about $0.032 \text{ }^{\circ}\text{C}$ per year over 2006–2025 – roughly $0.32 \text{ }^{\circ}\text{C}$ per decade. **Intercept:** $a_0 = -62.93 \text{ }^{\circ}\text{C}$ is the y-value the line would take at $x = 0$ (year 0). It has no physical meaning on its own here because the data only covers years far from zero; it is simply the value needed to fix the line's vertical position given the fitted slope and the data's mean point.

Goodness of fit: $r^2 = 0.788$ indicates that about 79% of the year-to-year variability in the temperature anomaly is explained by the linear trend with year; the remaining 21% is scatter from other factors (e.g., El Niño/La Niña cycles, volcanic aerosols, solar variability). The standard error of the estimate, $s_{y/x} = 0.0997 \text{ }^{\circ}\text{C}$, means that individual annual anomalies typically deviate from the regression line by about $\pm 0.10 \text{ }^{\circ}\text{C}$, which is a relatively small scatter given that the total warming over the 20-year span is roughly $0.55\text{--}0.6 \text{ }^{\circ}\text{C}$ – so the line captures the dominant trend well while leaving realistic year-to-year noise.

Residuals: The residual plot shows points scattered on both sides of zero with no strong curved (systematic) pattern, which supports the choice of a straight-line model. There is mild clustering of negative residuals in the mid-2010s to low-2020s and positive residuals at the start and the very end of the record, consistent with known short-term effects such as strong El Niño years (e.g., 2016, 2023–2024) pushing the anomaly briefly above the long-term linear trend, and La Niña-influenced years pulling it below. This is normal, physically-explainable scatter rather than evidence that the linear model is the wrong shape.